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INTERNAL COMPONENTS OF A COMPUTER SYSTEM

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INTERNAL COMPONENTS OF A COMPUTER SYSTEM

Objectives:

- **Identify the commercial hardware components of a computer system**, including the case, motherboard, chipset, RAM, microprocessor, graphics card, and more.
- **Develop a solid understanding of the fundamental concepts related to computer hardware**.
- **Detect potential incompatibilities** between the different components of a computer system.
- **Understand how magnetic storage devices work**, including their mechanical and electronic components, as well as their main characteristics.
- **Identify SSD storage devices**, including both traditional SSDs and **M.2/NVMe** drives.

INTERNAL COMPONENTS OF A COMPUTER SYSTEM (II)

- Connectors
- The Computer Chassis (Case)
- The Motherboard
- RAM Memory
- Graphics Card
- Microprocessor
- Expansion Cards
- Storage Devices
- Peripherals

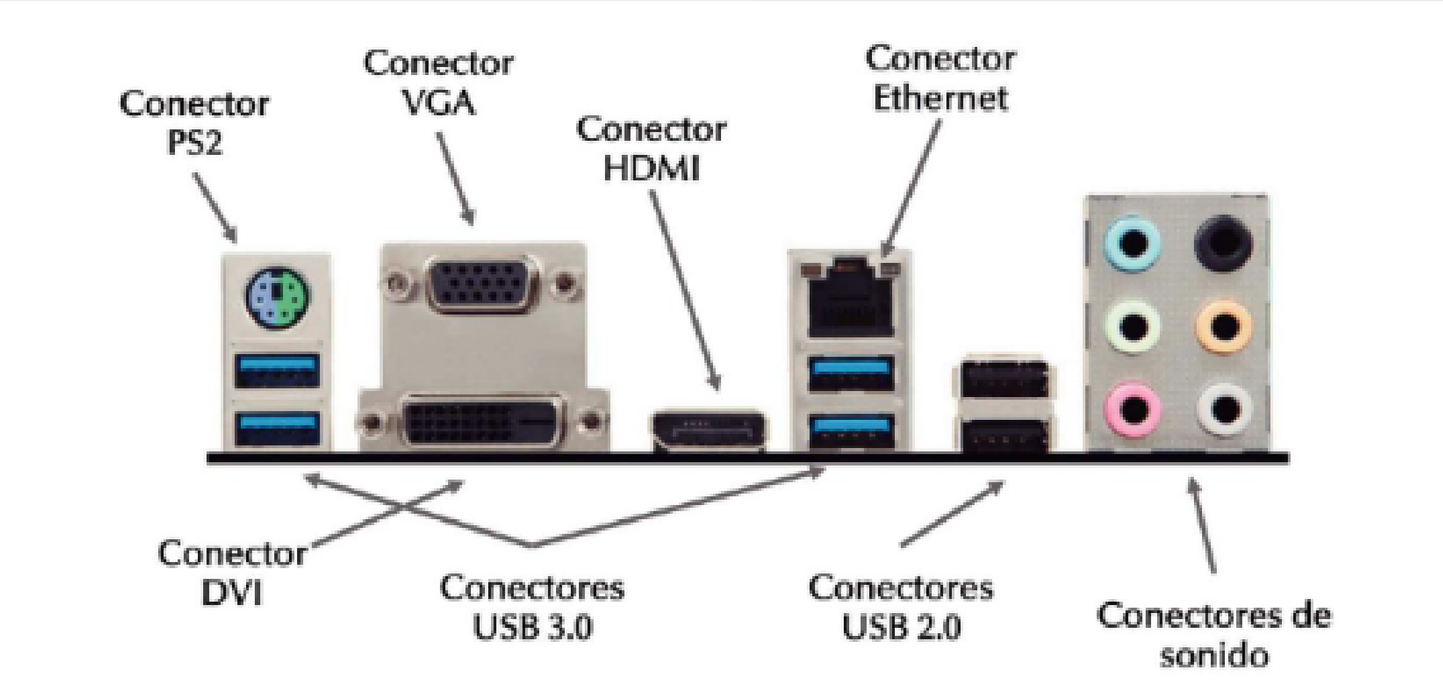
CONNECTORS

- Every computer system has **internal** and **external** connectors.

- Connectors are interconnection elements that enable communication between a computer’s internal components and external devices, such as peripherals, which add functionality to the system.

- All connectors are standardized. They may be used to connect the computer to the Internet, a mouse, a keyboard, speakers, and other external devices, or internally to connect components such as a hard disk drive, SSD, or optical drive.

CONNECTORS (II)



EXTERNAL: USB CONNECTOR

USB is one of the most widely used connectors due to its simplicity, durability, and reliability. Some of its main features are listed below:

- **Plug and Play.** All USB devices support this feature, meaning that devices can simply be connected and used without requiring manual configuration.

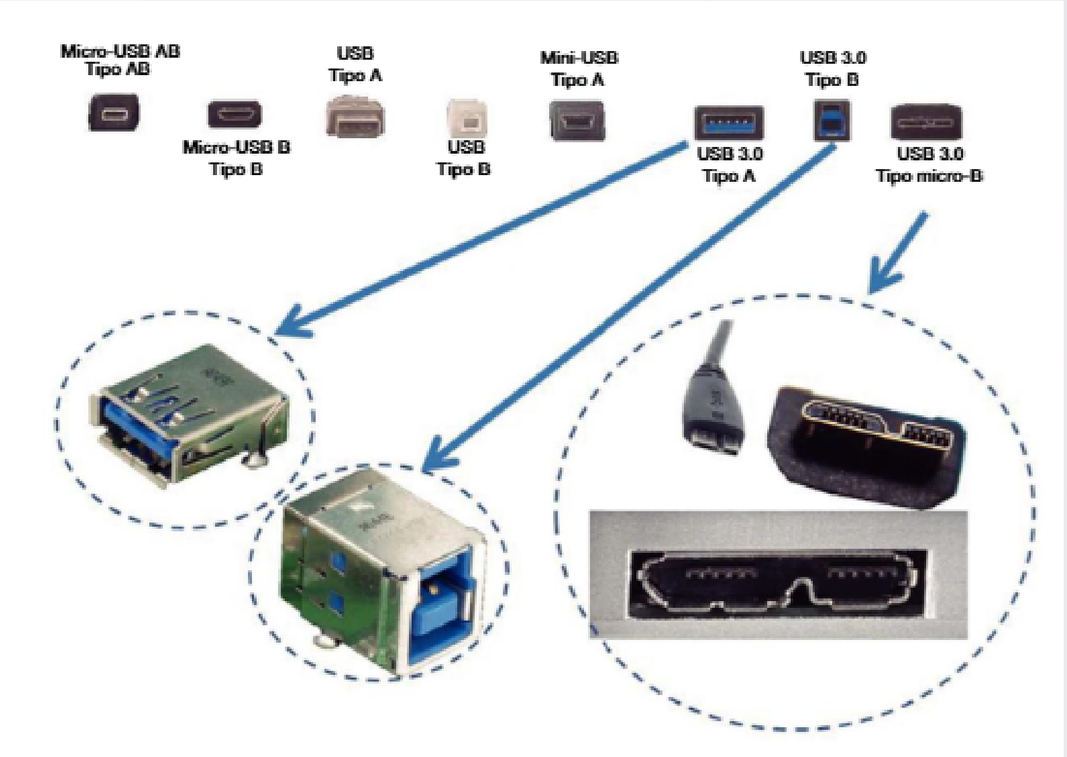
- **Speed.** USB 3.0 is ten times faster than its predecessor, USB 2.0, which supports speeds of up to 480 Mbps.

- **Backward Compatibility.** USB devices are backward compatible. This means that if a USB 3.0 device is connected to a USB 2.0 port, it will function correctly but not at its maximum speed. Likewise, a USB 2.0 device can be connected to a USB 3.0 port and will operate without issues, although it cannot take advantage of the additional performance offered by USB 3.0.

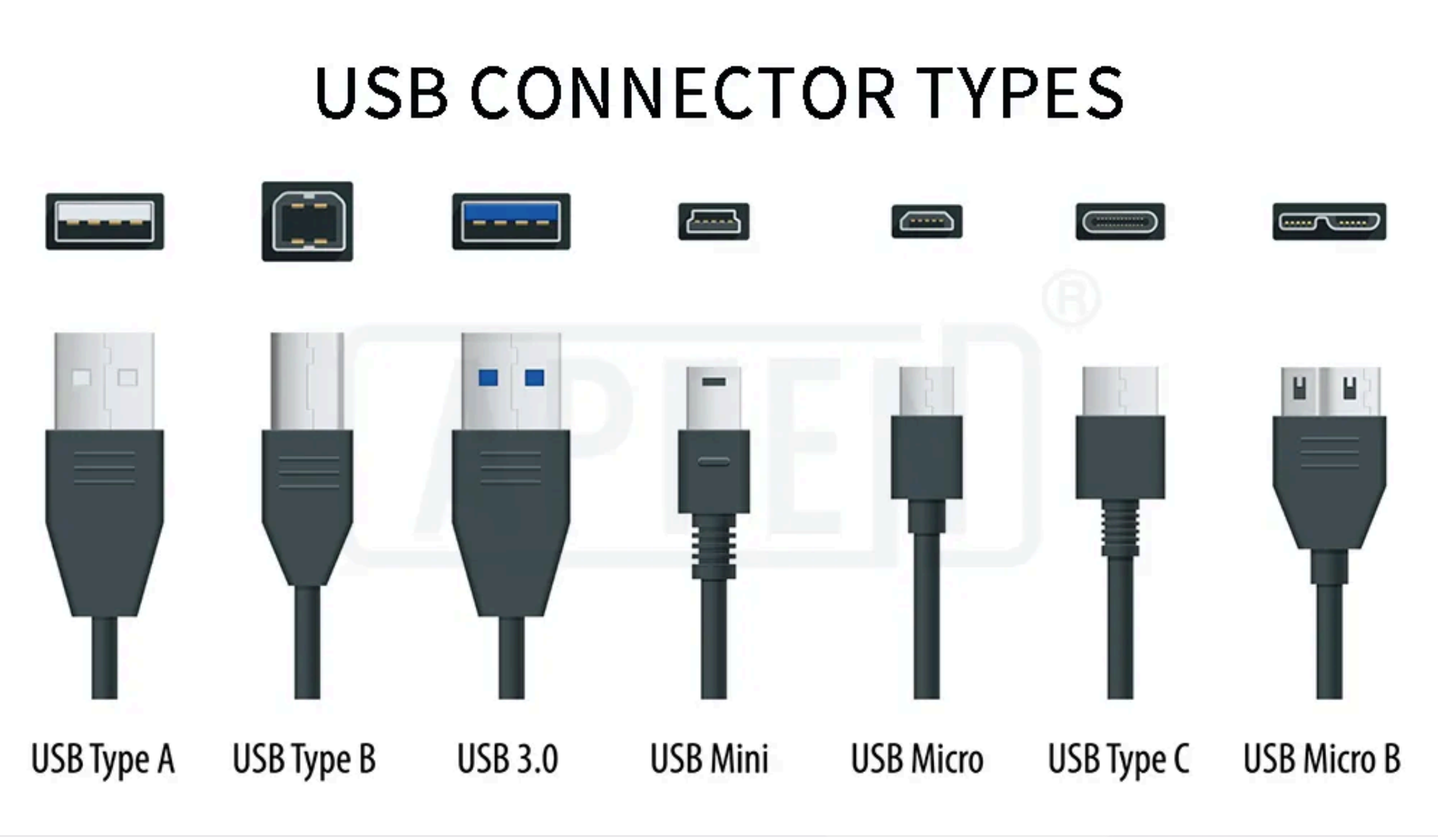
EXTERNAL: USB CONNECTOR (II)

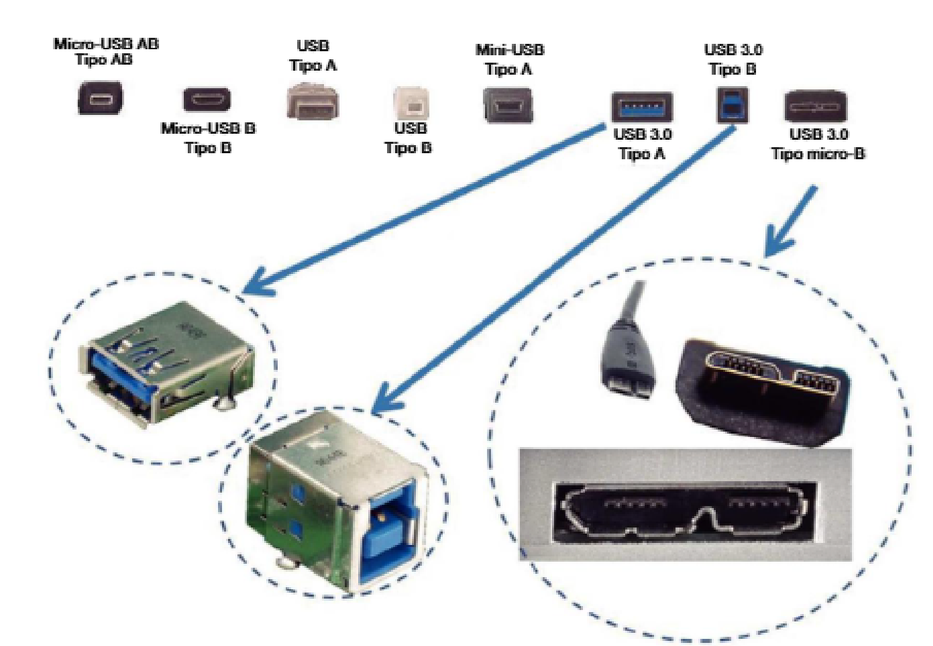
Main features (II):

- **Connection Types.** As shown in figure, there are several types of USB connectors. Type A and Type B connectors are similar and backward compatible with their USB 2.0 counterparts, while USB 3.0 introduced a new Micro-B connector. This connector is commonly used in devices such as external hard drives.



EXTERNAL: USB CONNECTOR (III)





EXTERNAL: USB CONNECTOR (IV)

Main features (III):

- USB 3.1. SuperSpeed USB. The new USB specification is USB 3.1, or SuperSpeed USB 10 Gbps, which provides a 10 Gbps transfer mode thanks to improved and more efficient data encoding. This specification offers twice the speed of USB 3.0 and is fully backward compatible. In addition to improving performance, it also increases energy efficiency, resulting in lower power consumption.

Think and answer: Is the USB 3.0 Type-A connector faster or slower than the USB 3.0 Micro-USB connector?

EXTERNAL: THUNDERBOLT CONNECTOR

Thunderbolt is a communications port found on Apple computers (MacBooks, iMacs, etc.) and is capable of transmitting video, audio, and data.

Intel invented the Thunderbolt port, just as it did the USB port.

The speed of this port lies in the fact that it uses an optical connection and, therefore, transmits data through light pulses rather than electrical pulses, resulting in higher transfer speeds.

EXTERNAL: THUNDERBOLT CONNECTOR (II)

Any peripheral device can be connected to this port, but due to its high speed, it is generally intended for **high-performance devices** such as a high-definition monitor, an external hard drive, and similar equipment.

Devices can be connected in a **daisy chain**, allowing up to six devices to be connected in this way. For example, in the previous image, the laptop could access the external hard drive even if the drive were not connected directly to the laptop, but indirectly through another device.

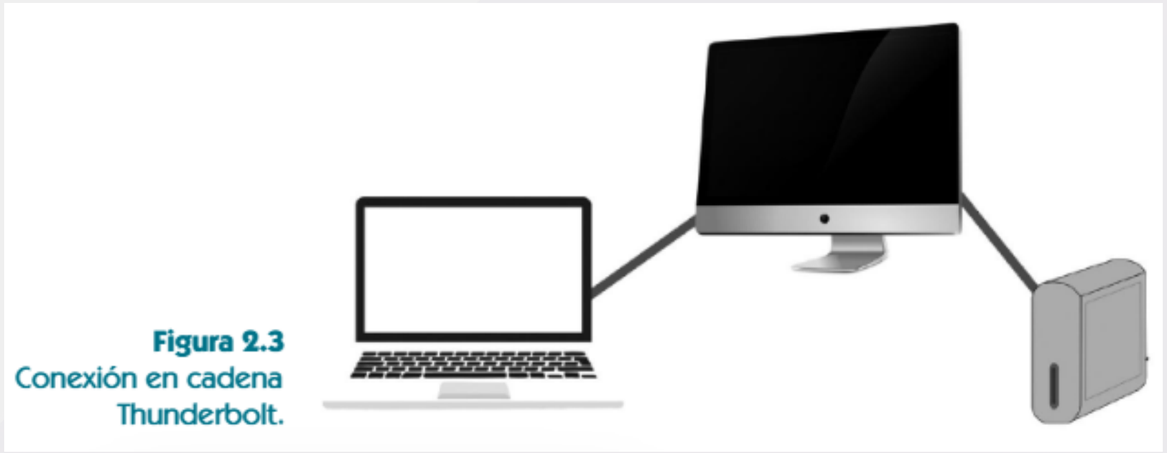
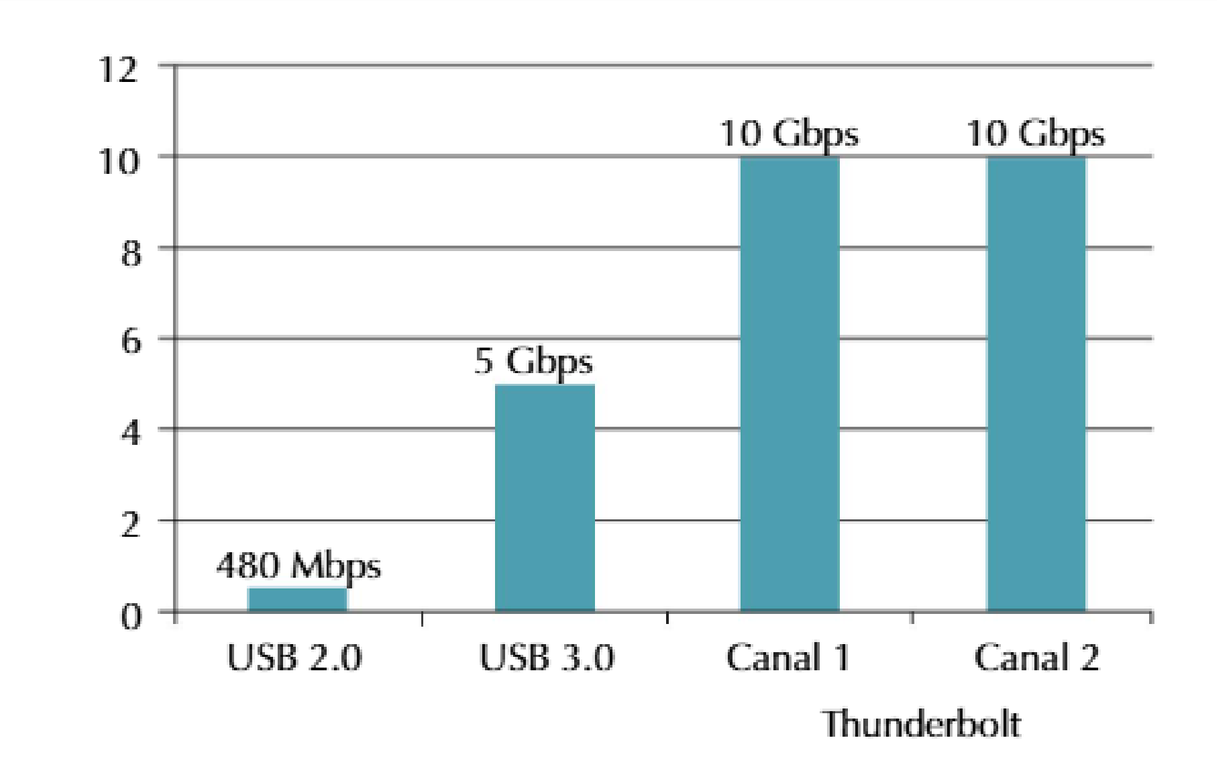


Figura 2.3
Conexión en cadena Thunderbolt.

EXTERNAL: THUNDERBOLT CONNECTOR (III)

This port, as shown in figure, is four times faster than the USB 3.0 port, as it operates using two simultaneous channels.



EXTERNAL: AUDIO CONNECTORS

Most devices normally include an **external audio jack**, which sometimes also functions as a microphone input.



On any motherboard, it is common to find at least three audio ports:

- **Green:** Line-out. Speakers.
- **Blue:** Line-in. It is not amplified like the microphone input.
- **Pink:** Microphone input.

Motherboards with 5.1 surround sound also include three additional connectors:

- **Gray:** Line-out for front speakers.
- **Black:** Line-out for rear speakers.
- **Orange:** Line-out for the subwoofer (bass speaker) or center speaker.

EXTERNAL: VGA, DVI AND HDMI PORTS FOR THE MONITOR

- The **VGA (Video Graphics Array)** port is the oldest of the three and, as can be seen, has 15 pins (connections). It is **analog**, unlike DP and HDMI, which are digital. This connector is usually blue.

- **DisplayPort (DP):** DisplayPort is a digital audio and video interface designed to connect computers to monitors and other display devices. It provides higher bandwidth than older standards such as VGA and DVI, allowing it to support high resolutions, high refresh rates, and multi-monitor configurations through a single connection. DisplayPort can also carry audio signals.

- The **HDMI (High Definition Multimedia Interface)** connector is more advanced than DVI and can transmit both video and audio at speeds of up to **5 Gbps**.

¿ANY QUESTION?